THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 6: **Triangles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $DE \parallel BC$ in $\triangle ABC$, $AB = 5$ cm, $AD = 3$ cm, $AC = 10$ cm, then EC (cm) is: 1

(a) 5 **(b)** 3 **(c)** 4 **(d)** 6

2. The areas of two similar triangles are 36 cm² and 100 cm². The ratio of their perimeters is 3: 1

(a) 6 **(b)** 7 **(c)** 5 **(d)** 4

3. The altitude to the hypotenuse of a right triangle divides it into 4 cm and 9 cm. The altitude (cm) is: 1

(a) 5 **(b)** 8 **(c)** 7 **(d)** 6

4. A pole 6 m high casts a shadow 4 m long. A tower casting a shadow 24 m has height (m): 1

(a) 35 **(b)** 37 **(c)** 38 **(d)** 36

5. The diagonals of a rhombus are 12 and 16. Its side squared is: 1

(a) 99 **(b)** 101 **(c)** 102 **(d)** 100

6. Two poles of heights 13 m and 8 m are 12 m apart. The distance between their tops (m) is: 1

(a) 13 **(b)** 12 **(c)** 14 **(d)** 15

7. If $DE \parallel BC$ in $\triangle ABC$, $AB = 7$ cm, $AD = 4$ cm, $AC = 21$ cm, then EC (cm) is: 1

(a) 9 **(b)** 10 **(c)** 8 **(d)** 11

8. The areas of two similar triangles are 36 cm² and 441 cm². The ratio of their perimeters is 2: 1

(a) 9 **(b)** 8 **(c)** 7 **(d)** 6

9. The altitude to the hypotenuse of a right triangle divides it into 3 cm and 12 cm. The altitude (cm) is: 1

(a) 8 **(b)** 7 **(c)** 6 **(d)** 5

10. A pole 8 m high casts a shadow 5 m long. A tower casting a shadow 40 m has height (m): 1

(a) 65 **(b)** 63 **(c)** 66 **(d)** 64

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 5$ cm, $AD = 3$ cm and $AC = 10$ cm, find AE and EC. 2

12. Two similar triangles have perimeters 21 cm and 42 cm. If a side of the smaller is 4 cm, find the corresponding side of the larger and the ratio of their areas. 2

13. In $\triangle ABC$, D and E are the midpoints of AB and AC and $BC = 8$ cm. Find DE and the ratio of areas of $\triangle ADE$ and $\triangle ABC$. 2

14. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 10$ cm, $AD = 4$ cm and $AC = 20$ cm, find AE and EC. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 9$ cm, $AD = 4$ cm and $AC = 18$ cm, find AE and EC. 3

16. $\triangle ABC \sim \triangle PQR$ with $AB = 3$ cm, $BC = 4$ cm, $CA = 5$ cm and $PQ = 6$ cm. Find QR and RP. 3

17. In a right triangle, the altitude to the hypotenuse divides it into segments of 4 cm and 9 cm. Find the altitude. 3

OR

In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 9$ cm, $AD = 2$ cm and $AC = 27$ cm, find AE and EC. 3

18. $\triangle ABC \sim \triangle PQR$ with $AB = 5$ cm, $BC = 6$ cm, $CA = 7$ cm and $PQ = 15$ cm. Find QR and RP. 3

OR

In a right triangle, the altitude to the hypotenuse divides it into segments of 2 cm and 8 cm. Find the altitude. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two poles of heights 14 m and 2 m stand on level ground 9 m apart. Find the distance between their tops. 5

OR

The diagonals of a rhombus are 12 cm and 16 cm. Find its side. 5

20. Two poles of heights 20 m and 8 m stand on level ground 16 m apart. Find the distance between their tops. 5

OR

The diagonals of a rhombus are 10 cm and 24 cm. Find its side. 5

END OF QUESTION PAPER — Total: 40 Marks

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